

HDOC Day-26

Right-aligned Patterns Using Nested Loops in C

Problem Statement

Write a menu-driven C program using switch-case and nested loops without arrays. Implement each pattern in a separate parameterless void function: right-aligned increasing asterisk triangle, right-aligned increasing number triangle, and right-aligned decreasing asterisk triangle. Repeat the menu until Exit is selected.

Algorithm

1. Start and display menu choices 1-4.
2. Read the menu choice and use switch-case.
3. For choices 1-3, call the corresponding parameterless function.
4. Inside the function, read n and repeat input until n is positive.
5. Choice 1: for i=1 to n, print n-i leading spaces and i stars.
6. Choice 2: for i=1 to n, print n-i leading spaces and numbers 1 to i.
7. Choice 3: for i=n down to 1, print n-i leading spaces and i stars.
8. Choice 4 exits; any other choice displays an error.
9. Stop.

Test Case Table

TC	Choice	n	Purpose	Expected Result
1	1	4	Valid	Right-aligned rows of 1,2,3,4 stars
2	2	4	Valid	Right-aligned rows: 1; 1 2; 1 2 3; 1 2 3 4
3	3	4	Valid	Right-aligned rows of 4,3,2,1 stars
4	1	1	Boundary	One star
5	2	0 then 3	Invalid then valid	Error, re-input, then 3-row pattern
6	9	-	Invalid choice	Error and menu repeats
7	4	-	Exit	Program terminates

C Program

```
#include <stdio.h>
```

```
void increasingStarTriangle(void);  
void increasingNumberTriangle(void);  
void decreasingStarTriangle(void);  
int readRows(void);
```

```
int main(void)  
{  
    int choice;
```

```

do
{
    printf("\n--- RIGHT-ALIGNED PATTERNS MENU ---\n");
    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1: increasingStarTriangle(); break;
        case 2: increasingNumberTriangle(); break;
        case 3: decreasingStarTriangle(); break;
        case 4: printf("Exiting program.\n"); break;
        default: printf("Invalid choice. Please try again.\n");
    }
} while (choice != 4);
return 0;
}

int readRows(void)
{
    int n;
    do
    {
        printf("Enter number of rows: ");
        scanf("%d", &n);
        if (n <= 0)
            printf("Number of rows must be positive. Try again.\n");
    } while (n <= 0);
    return n;
}

void increasingStarTriangle(void)
{
    int n = readRows(), i, j;
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

void increasingNumberTriangle(void)
{
    int n = readRows(), i, j;
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}

void decreasingStarTriangle(void)
{
    int n = readRows(), i, j;
    for (i = n; i >= 1; i--)

```

```
{  
    for (j = 1; j <= n - i; j++) printf(" ");  
    for (j = 1; j <= i; j++) printf("* ");  
    printf("\n");  
}  
}
```

Compilation and Execution

```
gcc hdoc26.c -o hdoc26  
./hdoc26
```

Expected Outputs for n = 4

Choice 1

```
*  
* *  
* * *  
* * * *
```

Choice 2

```
1  
1 2  
1 2 3  
1 2 3 4
```

Choice 3

```
* * * *  
* * *  
* *  
*
```

Screenshots

Screenshot - Source Code / Compilation / output

```
#include <stdio.h>
void increasingStarTriangle(void);
void increasingNumberTriangle(void);
void decreasingStarTriangle(void);
int readRows(void);
int main(void)
{
    int choice;
    do
    {
        printf("\n--- RIGHT-ALIGNED PATTERNS MENU ---\n");
        printf("1. Right-aligned increasing triangle\n");
        printf("2. Right-aligned increasing number triangle\n");
        printf("3. Right-aligned decreasing triangle\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1: increasingStarTriangle(); break;
            case 2: increasingNumberTriangle(); break;
            case 3: decreasingStarTriangle(); break;
            case 4: printf("Exiting program.\n"); break;
            default: printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 4);
    return 0;
}

int readRows(void)
{
    int n;
    do
    {
```

```
        int n;
        do
        {
            printf("Enter number of rows: ");
            scanf("%d", &n);
            if (n <= 0)
                printf("Number of rows must be positive. Try again.\n");
        } while (n <= 0);
        return n;
    }
}

void increasingStarTriangle(void)
{
    int n = readRows(), i, j;
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

void increasingNumberTriangle(void)
{
    int n = readRows(), i, j;
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}

void decreasingStarTriangle(void)
{
    int n = readRows(), i, j;
```

```

int n = readRows(), i, j;
for (i = n; i >= 1; i--)
{
    for (j = 1; j <= n - i; j++) printf(" ");
    for (j = 1; j <= i; j++) printf("* ");
    printf("\n");
}
}

```

```

(kali㉿kali)-[~/Desktop]
$ vi hdoc26.c

(kali㉿kali)-[~/Desktop]
$ gcc hdoc26.c -o hdoc26 -lm

```

```

(kali㉿kali)-[~/Desktop]
$ ./hdoc26

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 1
Enter number of rows: 4
  *
 * *
* * *
* * * *

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 2
Enter number of rows: 4
  1
 1 2
1 2 3
1 2 3 4

```

```

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 3
Enter number of rows: 4
* * * *
* * *
* *
*

```

```

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 1
Enter number of rows: 1
*

```

```

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 2
Enter number of rows: 2
1
1 2

```

```

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 9
Invalid choice. Please try again.

```

```

— RIGHT-ALIGNED PATTERNS MENU —
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit
Enter your choice: 4
Exiting program.

```